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A Project-Expo report on

"OPEN GENERATIVE AI - Web-Based AI Studio"

Submitted in partial fulfillment of the requirement in 6th semester for

REACT (BCSL657B)

Department of Computer Science & Engineering

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CERTIFICATE

Certified that the Project-Expo report on **“OPEN GENERATIVE AI - Web-Based AI Studio”,** carried out by **Chandu M (1AT23CS033), Digant S Hovale (1AT23CS045)** are bonafide students of Atria Institute of Technology, Bangalore, in partial fulfilment for the award of **Bachelor of Engineering in Computer Science & Engineering** of Visvesvaraya Technological University, Belgaum during the academic year 2025–2026. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library.

Prof. Harsha N	**Dr. Devi Kannan**	**Dr. Rajesha S**
Asst. Professor	**Professor and HOD**	**Principal**
Department of CSE Atria.I.T.	**Department of CSE Atria.I.T.**	**Atria.I.T.**

<u>DECLARATION</u>

We, Chandu M(1AT23CS033) & Digant S Hovale(1AT23CS045) students of 6th Semester Bachelor of Engineering, Department of Computer Science & Engineering, Atria Institute of Technology, Bangalore, hereby declare that the Project-Expo report titled **"OPEN GENERATIVE AI - Web-Based AI Studio",** has been carried out by us at Atria Institute of Technology, Bangalore, and submitted in partial fulfilment of the course requirement for the award of the degree of Bachelor of Engineering in Computer Science & Engineering of Visvesvaraya Technological University, Belagavi, during the academic year 2025–2026. We further declare that, to the best of our knowledge and belief, the work embodied in this report has not been submitted to any other university or institution for the award of any other degree

Place: Bangalore
Date: 13 May, 2026

<u>Signature of the student</u>

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Abstract

The rapid advancement of Generative Artificial Intelligence has revolutionized digital content creation, enabling the synthesis of images, videos, and lip-synchronized media from simple natural language prompts. However, the current ecosystem remains deeply fragmented — with individual capabilities scattered across dozens of proprietary platforms, each requiring separate subscriptions, accounts, and interfaces. This fragmentation creates significant friction for students, designers, and developers who require access to multiple generative modalities within a single workflow. Open Generative AI is a unified, web-based AI studio developed to address this gap, consolidating access to over 200 state-of-the-art generative AI models into a single, cohesive application. Built using Next.js 14, React 19, Tailwind CSS, Vite, and Electron, the platform offers four dedicated creative studios — ImageStudio, VideoStudio, LipSyncStudio, and WorkflowStudio — accessible through both a browser interface and a cross-platform desktop application.

The application is architected around React's core principles, serving as a comprehensive real-world demonstration of the VTU laboratory curriculum for BCSL657B. A modular, component-driven design ensures high reusability and clean separation of concerns, with shared child components propagating data through props following React's unidirectional data flow model. State management is handled exclusively through React Hooks — useState for generation lifecycle tracking and array-based history management, useEffect for side effects and API polling, and useRef for stable timer references. The technical centerpiece of the project is a two-step asynchronous polling architecture: upon generation request, a POST call submits the prompt and receives a request_id, after which periodic GET requests poll the result endpoint at three-second intervals until the AI model returns a completed output — handling the time-indeterminate nature of generative AI jobs entirely through React's reactive state system. All form inputs are implemented as controlled components with client-side validation, ensuring API keys, reference media, and prompt inputs are verified before any request is dispatched.

The user interface adopts a modern glassmorphism design language, implemented through Tailwind CSS utility classes with dynamic class application driven by React state — providing seamless visual feedback across loading, success, and error phases. The platform follows a Bring Your Own Key (BYOK) model, allowing users to supply their own API credentials to access 200+ models freely, without platform subscription costs. Packaged additionally as a desktop application via Electron, the project broadens accessibility beyond the browser without modifying the core React codebase. Overall, Open Generative AI successfully demonstrates that React's component model, Hook-based state management, and asynchronous integration patterns are powerful, industry-applicable tools — capable of underpinning a sophisticated, production-ready creative platform that genuinely democratizes access to state-of-the-art generative AI.

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1.Problem Statement

In today's rapidly evolving digital landscape, Generative AI has become a powerful tool for content creation — enabling users to produce images, videos, and lip-synced media using natural language prompts. However, the current ecosystem of generative AI tools is highly fragmented. Individual platforms such as Midjourney, Runway, Pika Labs, HeyGen, and Stable Diffusion each offer isolated capabilities behind separate accounts, subscriptions, and user interfaces.

This fragmentation creates significant friction for students, designers, content creators, and developers who need to work across multiple media types. Switching between platforms wastes time, disrupts creative workflows, and introduces inconsistency in quality and style. Furthermore, most existing tools are closed-source, expensive, and inaccessible to users in developing regions who cannot afford multiple SaaS subscriptions.

There is a clear need for a unified, open, and model-agnostic AI studio that consolidates access to the world's best generative AI models into a single, intuitive interface — without locking users into any one provider.

2.Objective

The primary objective of Open Generative AI is to design and develop a unified, web-based AI studio that provides seamless access to 200+ state-of-the-art generative AI models through a single platform.
Specific objectives include:

- Build a modular React-based application with dedicated studios for image generation, video generation, lip-sync, and workflow automation
- Implement a clean, reusable component architecture using React functional components and Hooks
- Enable users to bring their own API key (BYOK) to interface with AI providers — keeping access free and flexible
- Provide real-time generation feedback through asynchronous API polling using useEffect and setInterval
- Deliver a responsive, modern glassmorphism UI accessible on both browser and desktop (via Electron)
- Demonstrate all five React laboratory outcomes of VTU course BCSL657B in a real-world application

Vite

Vite is used as the development build tool for isolated component testing and rapid hot-module replacement (HMR) during development of individual studio modules.

Electron

The application is packaged as a cross-platform desktop application using Electron. The Next.js build output is served within an Electron BrowserWindow, allowing users to use the AI studio as a native desktop application without needing a browser — while still being powered by the same React codebase.

3.Existing System

The existing ecosystem consists of several standalone, purpose-specific AI generation platforms:

Platform	Capability	Limitation
Midjourney	Image generation	Discord-only interface, paid
Runway ML	Video generation	Expensive subscription
HeyGen	Lip-sync / Avatar	Enterprise pricing
Pika Labs	Image-to-video	Limited free tier
Stable Diffusion (local)	Image generation	Requires powerful GPU hardware
Leonardo AI	Image generation	Model-locked, proprietary

Drawbacks of the Existing System:

No unified interface — users must maintain accounts across 5–10 platforms

High cost — most platforms charge $15–$100/month per service

No cross-media workflow — impossible to chain image → video → lip-sync in one place

Vendor lock-in — users are restricted to models chosen by the platform

Limited transparency — closed-source; users cannot inspect or customize model parameters

Poor accessibility — most platforms are unavailable or unaffordable in developing countries

4.Proposed System

Open Generative AI proposes a single, unified web-based AI studio that overcomes all limitations of the existing fragmented ecosystem.

Key features of the proposed system:

- Unified Studio Interface — One platform with four dedicated creative studios: ImageStudio, VideoStudio, LipSyncStudio, and WorkflowStudio
- 200+ Model Access — Integrates with a wide external AI API aggregator, giving users access to models from multiple AI providers through one API key
- BYOK (Bring Your Own Key) — Users supply their own API key, eliminating platform subscription costs
- Two-Step Async Polling — Handles long-running AI generation jobs through a submit-then-poll architecture, providing real-time status updates
- Generation & Upload History — Session-persistent history of all generated outputs with download and delete capabilities
- Cross-Studio Workflow — WorkflowStudio enables chaining of generation steps (e.g., generate image → animate → lip-sync)
- Desktop App — Electron wrapper packages the application as a native desktop app for Windows, macOS, and Linux
- Modern UI — Glassmorphism design with full responsiveness built using Tailwind CSS

Comparison:

Feature	Existing System	Proposed System
Number of models	1–10 per platform	200+
Platforms needed	5–10	1
Monthly cost	\$50–\$300+	Free (BYOK)
Cross-media workflow	Not supported	Fully supported
Desktop support	Browser only	Browser + Electron
Open & customizable	No	Yes

5.Functional Requirements

FR1 — User Authentication & API Key Management

- The system shall display an ApiKeyModal when a user attempts to generate content without a stored API key
- The system shall validate that the API key field is non-empty before saving
- The system shall allow the user to update or revoke the API key at any time

FR2 — Image Generation (ImageStudio)

- The system shall accept a text prompt and optional reference image as input
- The system shall allow selection of an AI model from a dropdown of 200+ available models
- The system shall support configuration of aspect ratio, image count, and style parameters
- The system shall display generation status (loading, success, error) in real-time

FR3 — Video Generation (VideoStudio)

- The system shall support both text-to-video and image-to-video generation modes
- For image-to-video models, the system shall require at least one reference image upload
- The system shall support upload of up to 14 reference images with dynamic count tracking
- The system shall allow configuration of video duration and motion parameters

FR4 — Lip-Sync Generation (LipSyncStudio)

- The system shall require both a reference face media file and an audio file before submission
- The system shall validate file formats for audio (mp3, wav, m4a) and video/image inputs
- The system shall display the lip-synced video output upon completion

FR5 — Workflow Automation (WorkflowStudio)

- The system shall allow users to define multi-step AI pipelines
- Each pipeline step shall pass its output as input to the next step
- The system shall display intermediate results at each pipeline stage

FR6 — History Management

- The system shall maintain a session-level generation history array for each studio
- The system shall support deletion of individual history items
- The system shall support full history reset (clear all)

FR7 — Drag-and-Drop File Upload

- The system shall support drag-and-drop upload of image, video, and audio files
- The system shall enforce file type and count limits with appropriate error messages

FR8 — Asynchronous API Polling

- The system shall submit generation requests via HTTP POST and receive a request_id
- The system shall poll the result endpoint via HTTP GET at 3-second intervals
- The system shall terminate polling upon receiving a completed or failed status

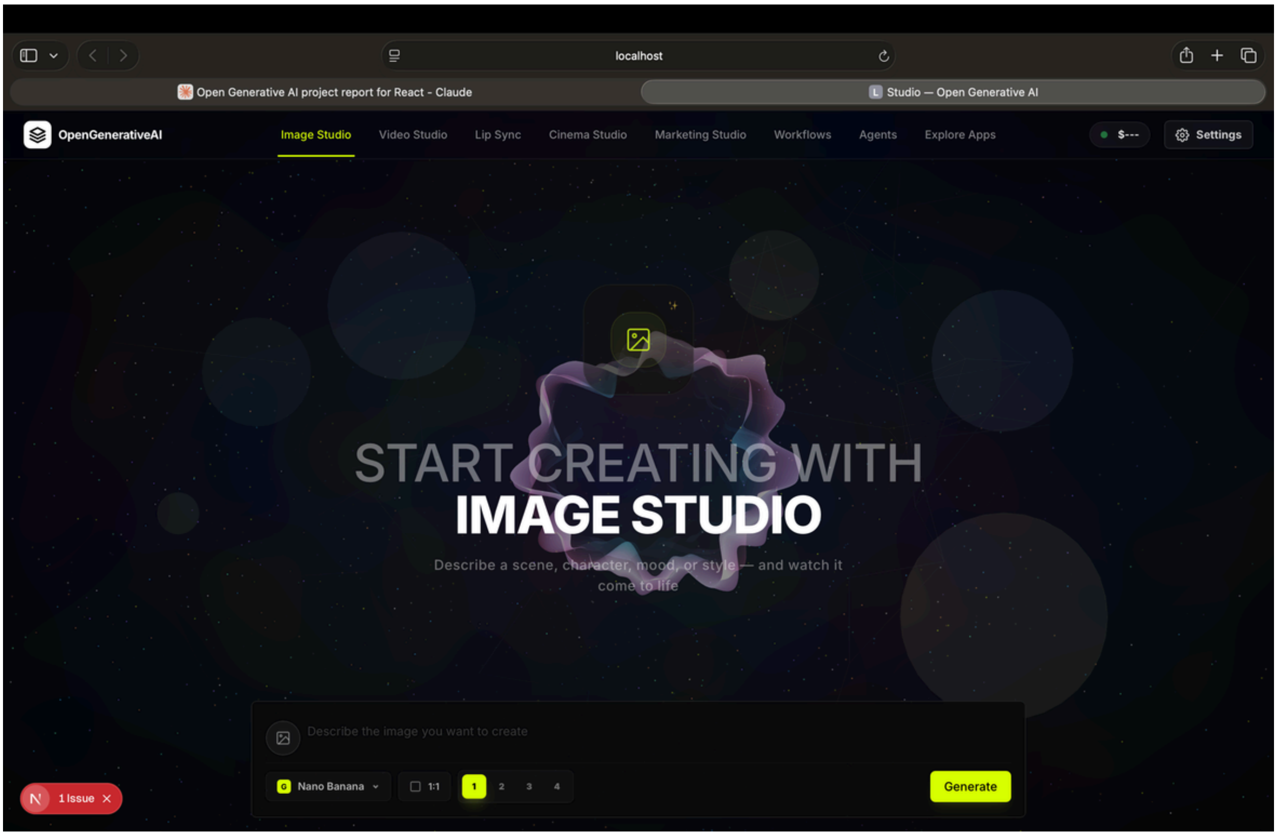
6.Tools & Technologies

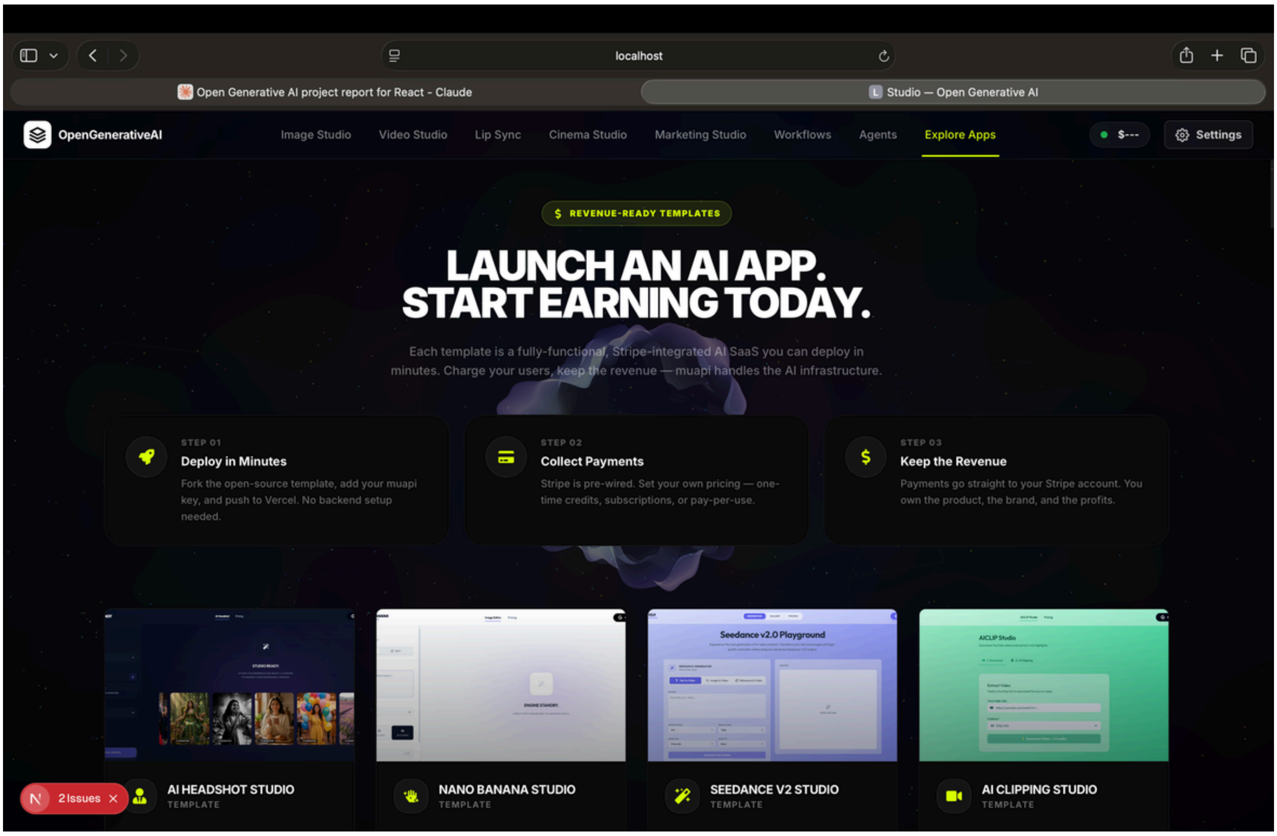
Category	Technology	Purpose
UI Library	React 19	Component-based UI development
Framework	Next.js 14 (App Router)	Full-stack React framework, routing
Styling	Tailwind CSS	Utility-first responsive UI, glassmorphism
Build Tool	Vite	Fast HMR and component bundling
Desktop	Electron	Cross-platform desktop app wrapper
Language	JavaScript (ES2023)	Core application logic
State Management	React Hooks (`useState`, `useEffect`, `useRef`)	Local component state management
API Communication	Fetch API	HTTP requests to AI model endpoints
Version Control	Git + GitHub	Source code management and collaboration
Runtime	Node.js v18+	JavaScript server-side runtime
Package Manager	npm	Dependency management
Editor	VS Code	Development environment

7.Expected Outcome

Upon successful completion of the project, the following outcomes are expected:
A fully functional web application accessible via browser and as a desktop app, offering image, video, and lip-sync generation from a single unified interface
Practical demonstration of all React course outcomes — functional components, Hooks, prop-based data flow, controlled forms, and real-time API-driven state updates — as specified in VTU BCSL657B
Real-world asynchronous architecture — a working two-step polling system that reliably handles AI generation jobs ranging from 5 seconds to several minutes in duration
Production-grade UI/UX — a responsive, glassmorphism-styled interface that is intuitive for both technical and non-technical users
Model-agnostic platform — users are not locked into any single AI provider; access to 200+ models gives creative flexibility unmatched by any single existing platform
Cost-accessible tool — by adopting a BYOK (Bring Your Own Key) model, the application remains free to use for students and independent creators
Reusable codebase — the modular component architecture ensures the project can be extended with additional studios (e.g., AudioStudio, 3D Studio) with minimal refactoring

8.ScreenShots





9.Geotag Photos